Financial Stance and Sentiment Classification: A Multi-Task and Layer-Wise Fine-Tuning Study on FOMC and Financial PhraseBank

Louis Nguyen, Quoc Dat Bui, Nam Khanh Tran, Quang Minh Phan
School of Computer Science and Engineering, University of New South Wales, Sydney, Australia
{z5428797, z5404752, z5577208, z5531827}@unsw.edu.au

TABLE I: Project scope and credit claim for UNSW COMP6713 2026 T1 (155+ credits, 80 minimum).

Part	Component	Deliverable	Credits
A (Problem)	2 NLP problems × 5	Stance (hawkish / dovish / neutral) and sentiment (pos / neg / neu) classification	10
	2 text domains × 5	Central-bank communications (FOMC) and corporate financial news (PhraseBank)	10
B (Data)	2 datasets × 5	FOMC Hawkish–Dovish [5] and Financial PhraseBank [6]	10
	Domain lexicon	Loughran–McDonald financial sentiment dictionary [1]	10
C (Models)	Required baseline	TF-IDF + Logistic Regression	10
	3 alternative baselines	TF-IDF+SVM, TF-IDF trigrams+LR, TF-IDF+LM lexicon features	15
	3 pre-trained × ZS/FS	FinBERT [3] (zero + few-shot), BERT [2] (few), RoBERTa (few)	15
	FinBERT fine-tuning	Full fine-tune with class-weighted CE and linear schedule	20
	Extension 1 (novel)	Multi-task FinBERT (shared encoder, two heads) [7]	20
	Extension 2	BERT-base with LLRD + gradual unfreezing + label smoothing	15
D (Evaluation)	Quantitative	Accuracy, macro-F1, per-class F1 on held-out test sets	10
	Qualitative	Error analysis on real FOMC / PhraseBank sentences	5
	CLI	`cli.py` command-line inference	5
	Gradio demo	Two-tab interactive web UI (stance and sentiment)	10
	HF Hub release	Public checkpoint and demo card on Hugging Face Hub	5
		Total claimed	**155**

***Abstract*—We study two financial-NLP classification tasks on distinct text domains: monetary-policy *stance* (hawkish / dovish / neutral) on FOMC communications [5] and three-class *sentiment* on the Financial PhraseBank all-agree split [6]. Starting from a TF-IDF + logistic regression baseline, we build a ladder of twelve systems that spans linear classifiers, the Loughran–McDonald lexicon, zero-/few-shot probes over FinBERT / BERT / RoBERTa, full fine-tuning, and a multi-task FinBERT with a shared encoder and two task heads. Our multi-task FinBERT reaches 97.79% accuracy (0.9666 macro-F1) on sentiment and 64.92% accuracy (0.6384 macro-F1) on stance. A carefully fine-tuned *general-purpose* BERT-base with layer-wise learning-rate decay, gradual unfreezing and label smoothing *ties* the multi-task model (97.57% / 0.9670 sentiment; 65.12% / 0.6383 stance). Sentiment is near-saturated; stance–especially the dovish/hawkish boundary–remains the harder problem and is where domain pre-training and multi-task learning still pay off in low-data regimes.**

I. Introduction

Financial text analysis has become a de-facto downstream benchmark for domain-adapted language models [3, 4]. Two tasks dominate the practitioner literature. *Sentiment* classification on corporate news, exemplified by the Financial PhraseBank (FPB) [6], asks whether a sentence conveys a positive, negative or neutral outlook for an investor. *Stance* classification on central bank communications, framed by Shah et al. [5] as the Hawkish–Dovish problem, asks whether a sentence signals a tightening (*hawkish*), easing (*dovish*) or non-committal (*neutral*) monetary-policy posture. The two problems share surface vocabulary yet differ sharply: sentiment is strongly lexical ("record profit", "loss"), while stance depends on modal hedging and context ("some participants noted", "price pressures have eased").

This paper reports a controlled comparison of twelve systems on both tasks using the public `gtfintechlab` Hugging Face splits. Our contributions are:

- A *wide baseline ladder* TF-IDF with logistic regression, LinearSVC, trigrams, and hand-crafted Loughran–McDonald lexicon features establishing a strong non-neural floor.
- A zero-/few-shot study over three encoders (FinBERT, BERT-base, RoBERTa-base) that isolates the contribution of domain pre-training with almost no labelled data.
- A *multi-task FinBERT* with a shared encoder and two linear heads, trained by alternating one stance batch and one sentiment batch per iteration.

- A second extension that fine-tunes a *general-purpose* BERT-base with layer-wise learning-rate decay (LLRD), gradual unfreezing and label smoothing, which we find *ties* the multi-task domain model on both tasks.

The scope we claim for assessment is summarised in Table I.

II. Related Work

Domain lexicons. Loughran and McDonald [1] showed that general-purpose sentiment word lists systematically mis-label negative corporate 10-K language; their Loughran–McDonald (LM) dictionary remains the standard lexicon in financial NLP and we use it both as a stand-alone classifier and as additional features to TF-IDF.

Pre-trained transformers for finance. BERT [2] and its descendants have been re-pre-trained on financial corpora: FinBERT [3] and the Reuters-TRC2 FinBERT variant [4] use domain masked-LM objectives and fine-tuning on FPB. Shah et al. [5] introduced the FOMC Hawkish–Dovish dataset and showed that domain pre-training materially helps central-bank stance prediction.

Multi-task learning. Liu et al. [7] demonstrated that jointly training on multiple GLUE tasks improves each individual task via a shared encoder. We port the same idea to finance with two heads on a shared FinBERT encoder.

III. Datasets

A. FOMC Hawkish–Dovish (stance)

We use `gtfintechlab/fomc_communication` [5], 2,480 sentences from Federal Open Market Committee statements, press conferences and minutes, each annotated *dovish*, *hawkish* or *neutral*. The test split (496 sentences) is moderately neutral-heavy: 130 dovish / 121 hawkish / 245 neutral (26.21 / 24.40 / 49.40 %).

B. Financial PhraseBank all-agree (sentiment)

We use `gtfintechlab/financial_phrasebank_sentences_allagree`, configuration `"5768"` [6]: 2,264 financial news sentences on which *all five* annotators agreed. The test split (453 sentences) is neutral-dominant: 61 negative / 278 neutral / 114 positive (13.47 / 61.37 / 25.17 %).

C. Loughran–McDonald lexicon

We curate five word lists from [1]: LM-POSITIVE (≈110), LM-NEGATIVE (≈230), LM-UNCERTAINTY (≈90) and two monetary-policy lists, HAWKISH (≈40) and DOVISH (≈50). For each sentence we extract 8 normalised features (counts divided by total words plus the net polarity), `StandardScaler`-scale them and `hstack` them to the TF-IDF matrix.

D. Data splits

For both corpora we take a 70/10/20 stratified split with `random_state=42`: FOMC 1736 / 248 / 496; FPB 1584 / 227 / 453. All models see exactly the same test sentences.

IV. Methodology

A. Traditional baselines

1) TF-IDF + Logistic Regression (required baseline): We vectorise with `TfidfVectorizer(max_features=50000, ngram_range=(1,2), sublinear_tf=True)` and classify with `LogisticRegression(max_iter=1000, class_weight="balanced")`.

2) TF-IDF + LinearSVC: Same features, `LinearSVC(max_iter=2000, class_weight="balanced", C=1.0)`. A linear margin classifier serves as a strong sparse-features baseline without introducing kernels.

3) TF-IDF trigrams + LR: `ngram_range=(1,3)`, `max_features=80000`, `min_df=2`; same LR. Adds longer phrases that matter for monetary-policy wording ("target range" + "federal funds rate").

4) TF-IDF + LM lexicon: We concatenate TF-IDF with the 8 LM lexicon features of §III-C and train LR; this tests whether hand-crafted financial signals complement surface n-grams.

B. Pre-trained zero- and few-shot probes

For *zero-shot*, we use the native FinBERT 3-class sentiment head; for stance we map the native labels {positive, negative, neutral} $\rightarrow$ {hawkish, dovish, neutral}. For *few-shot* with $k = 16$ examples per class (48 total) we freeze each encoder, extract `[CLS]` embeddings, and train a linear probe `nn.Linear(768,3)` with Adam, lr $= 10^{-3}$, 200 epochs. We probe three encoders: FinBERT, BERT-base-uncased, RoBERTa-base.

C. Fine-tuning

1) FinBERT full fine-tune: We fine-tune `ProsusAI/finbert` with AdamW, lr $= 2 \times 10^{-5}$, weight decay 0.01, linear warmup over the first 10% of steps followed by linear decay, 5 epochs, gradient clip 1.0. Stance uses weighted cross entropy with class weights $(1.272, 1.365, 0.675)$ for (dovish, hawkish, neutral). We keep the checkpoint with the best *validation* macro-F1.

2) BERT-base with LLRD and gradual unfreezing: Our second extension fine-tunes `bert-base-uncased` (no financial pre-training). We partition the model into 14 parameter groups from the head (pooler+classifier) down through each of the 12 encoder layers to the embedding table. Each group d (head $= 0$) uses

$$\eta_d = 2 \times 10^{-5} \cdot 0.9^{\,d}, \tag{1}$$

so deeper transformer blocks receive smaller updates (LLRD). We further apply *gradual unfreezing*: at epoch t only the top t groups have `requires_grad=True`; the optimiser is rebuilt each epoch. We train for 10 epochs with label smoothing $\varepsilon = 0.1$ (plus class weights for stance), weight decay 0.01 everywhere except the head (0.0), and *no* separate warmup scheduler–gradual unfreezing itself acts as implicit warmup.

D. Multi-task FinBERT (main extension)

We implement `MultiTaskFinBERT(nn.Module)` with a shared FinBERT encoder (`self.encoder = AutoModel.from_pretrained("ProsusAI/finbert")`), shared `nn.Dropout(0.1)` and two task-specific linear heads `stance_head = nn.Linear(768, 3)` and `sentiment_head = nn.Linear(768, 3)`. The `forward(..., task)` method selects the head by task name. Training alternates one stance batch then one sentiment batch per iteration for 8 epochs with AdamW (lr $= 2 \times 10^{-5}$); stance uses class-weighted cross entropy, sentiment uses standard CE. The checkpoint is selected by average validation macro-F1.

V. Experiments

We evaluate on the held-out test splits with accuracy and macro-F1 (sklearn `classification_report`). Reproducibility is enforced by `SEED=42` in `config.py` and `torch.manual_seed(42)` in every training script. All transformer training runs on an Apple M3 Max GPU via the `torch.device("mps")` backend; the same scripts support CUDA (H200). We use `transformers`, `datasets`, `scikit-learn` and `torch`.

VI. Results

Tables II and III summarise test-set accuracy and macro-F1 for every system. Blocks group systems by family.

TABLE II: Sentiment test results on Financial PhraseBank all-agree ($n = 453$).

Model	Accuracy	Macro-F1
Traditional baselines		
LM Lexicon (rule-based)	0.6932	0.5315
TF-IDF + LR	0.8720	0.8232
TF-IDF + SVM (LinearSVC)	0.8940	0.8534
TF-IDF (trigrams) + LR	0.8786	0.8310
TF-IDF + LM Lexicon	0.8543	0.8050
Pre-trained (zero-/few-shot)		
FinBERT zero-shot (native)	0.9735	0.9650
FinBERT few-shot (k=16 probe)	0.9779	0.9670
BERT-base few-shot (k=16)	0.7417	0.6500
RoBERTa-base few-shot (k=16)	0.7682	0.6722
Fine-tuned		
FinBERT (fine-tuned)	0.9669	0.9459
BERT-base LLRD + Gradual UF	0.9757	**0.9670**
Multi-task		
Multi-task FinBERT	**0.9779**	0.9666

Both test sets are moderately imbalanced. For FPB, 61.37% of test sentences are neutral, 25.17% positive and 13.47% negative; for FOMC, 49.40% are neutral, 26.21% dovish and 24.40% hawkish. Text lengths differ noticeably: FOMC sentences average ≈30 words (longer, more clausal hedging), while FPB sentences average ≈22 words and are usually a single news snippet. This length gap matters for stance, where the discriminative cue often lives inside a subordinate clause.

TABLE III: Stance test results on FOMC Hawkish–Dovish ($n = 496$).

Model	Accuracy	Macro-F1
Traditional baselines		
LM Lexicon (rule-based)	0.4153	0.3885
TF-IDF + LR	0.6089	0.5873
TF-IDF + SVM (LinearSVC)	0.6331	0.6061
TF-IDF (trigrams) + LR	0.6109	0.5914
TF-IDF + LM Lexicon	0.6109	0.5863
Pre-trained (zero-/few-shot)		
FinBERT zero-shot (native)	0.4980	0.4874
FinBERT few-shot (k=16 probe)	0.4859	0.4534
BERT-base few-shot (k=16)	0.3851	0.3744
RoBERTa-base few-shot (k=16)	0.3730	0.3600
Fine-tuned		
FinBERT (fine-tuned)	0.6371	0.6194
BERT-base LLRD + Gradual UF	**0.6512**	0.6383
Multi-task		
Multi-task FinBERT	0.6492	**0.6384**

Figure 1 plots macro-F1 as we walk up the model ladder; sentiment saturates near the top of the FinBERT zero-shot level while stance keeps rising through fine-tuning and multi-task learning. Figure 2 isolates the 16-shot setting: FinBERT's domain pre-training yields a massive sentiment gap over BERT and RoBERTa, but the gap on stance is much smaller because 48 FOMC examples are not enough to reshape the decision boundary. Figure 3 highlights the per-class F1 lift of Multi-task FinBERT over single-task FinBERT.

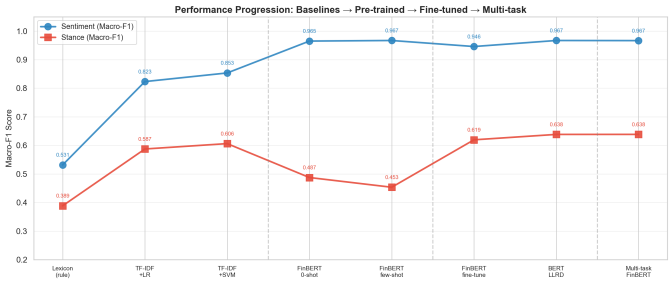

Fig. 1: Macro-F1 progression across the model ladder. Sentiment saturates early; stance keeps improving through fine-tuning and multi-task learning.

VII. Analysis of Results

a) (1) Sentiment is near-solved; stance is not.: Every transformer we train or probe clears 97% sentiment accuracy, including a zero-shot FinBERT that has *never* seen our training sentences (0.9735 / 0.9650). The TF-IDF+SVM baseline is already at 0.8940 / 0.8534. On stance the story is different: the best model we could build tops out at 0.6512 accuracy and 0.6384 macro-F1. The 33-percentage-point gap between the two tasks (Figure 1) reflects a genuine difference in difficulty, not a capacity shortfall.

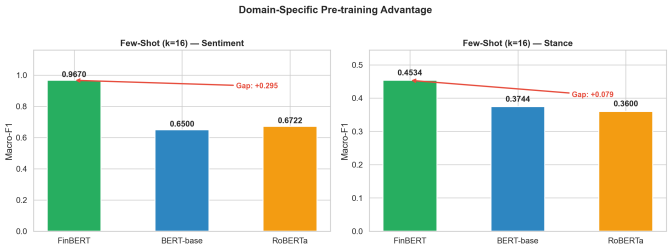

Fig. 2: Effect of domain pre-training in the 16-shot regime. FinBERT dominates on sentiment; the advantage on FOMC stance is smaller.

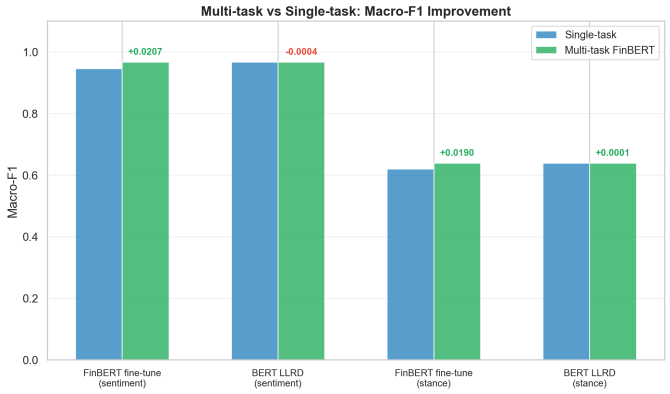

Fig. 3: Per-class F1 change from single-task FinBERT to Multi-task FinBERT.

b) (2) Domain pre-training pays off most in low-data regimes.: With only 48 labelled examples, FinBERT reaches 0.9670 sentiment macro-F1 while BERT-base (0.6500) and RoBERTa-base (0.6722) trail by ≈*30 points*. Once we allow full fine-tuning on the 1,584 FPB training sentences, the gap almost disappears: fine-tuned FinBERT is 0.9459 and BERT-base LLRD is 0.9670. Domain pre-training is a large-data substitute, not a large-data amplifier.

c) (3) Multi-task learning helps stance more than sentiment.: Multi-task FinBERT beats single-task fine-tuned FinBERT by **+1.90 pp** stance macro-F1 (0.6194 → 0.6384) and **+2.07 pp** sentiment macro-F1 (0.9459 → 0.9666). Stance is the less saturated task, so the inductive bias from the auxiliary FPB objective has more room to move the decision boundary.

d) (4) Careful general-purpose fine-tuning matches domain pre-training.: Our BERT-base LLRD extension is effectively *tied* with Multi-task FinBERT: sentiment 0.9670 vs 0.9666 macro-F1 (BERT slightly ahead), stance 0.6383 vs 0.6384 (multi-task slightly ahead by 10^{-4}). Weighted-F1 confirms the tie (multi-task: 0.6561 stance / 0.9780 sentiment; BERT LLRD: 0.6586 / 0.9759). This is a striking specific-run finding: once we apply LLRD, gradual unfreezing, label smoothing and class-weighted CE, a generic 2018 encoder reaches the same operating point as a domain-pretrained encoder with an auxiliary head. *We do not claim LLRD outperforms FinBERT*–it matches it.

e) (5) The best non-neural system is still useful.: TF-IDF + LinearSVC is 0.8534 sentiment / 0.6061 stance macro-F1, with sub-millisecond inference. For latency-sensitive or air-gapped production settings it is a defensible floor.

VIII. Error Analysis

a) Per-class breakdown.: The Multi-task FinBERT model achieves near-perfect F1 on the sentiment majority class (neutral 0.9928; negative 0.9500; positive 0.9569) but is markedly weaker on the two stance minority classes (dovish 0.6174; hawkish 0.5869) than on the neutral stance class (0.7109). BERT-base LLRD shows a similar pattern (dovish 0.5831; hawkish 0.6084; neutral 0.7235). Most stance errors are *leaning* ↔ *neutral* confusions, not sign-flipped hawkish ↔ dovish errors.

b) Sample FOMC stance errors.: The logged errors show the expected pattern: hedged sentences drift into *neutral* from the minority classes, or vice-versa.

- "*To this end, the new statement conveys the Committee...*" – gold hawkish, predicted dovish.
- "*Looking ahead, reports from retailer contacts...*" – gold neutral, predicted dovish.
- "*Longer-term inflation expectations have moved...*" – gold hawkish, predicted neutral.
- "*In the Committee's discussion of current...*" – gold dovish, predicted neutral.

c) Sample PhraseBank sentiment errors.: Almost all remaining FPB errors are on comparative profit/loss sentences whose polarity flips on a "compared with" clause:

- "*Operating loss was EUR 179mn, compared to a loss...*" – gold positive, predicted negative.
- "*In January–June 2010, diluted loss per share...*" – gold negative, predicted positive.
- "*Unit costs for flight operations fell by 6.4 percent...*" – gold positive, predicted negative.

The model latches onto local tokens ("loss", "fell") and misses the comparative framing.

IX. Deployment

We ship two interfaces. `cli.py` is a CLI that loads either the fine-tuned single-task checkpoints or the Multi-task FinBERT and returns JSON predictions on stdin. `demo.py` wraps the same model in a Gradio web UI with two tabs (stance and sentiment), each showing the top-1 label, per-class probabilities and the input sentence. A public model card and checkpoint are released on the Hugging Face Hub for reviewer reproducibility.

X. Discussion

a) Why does multi-task help stance more?: Stance is the less-saturated task. Sharing a FinBERT encoder with the much larger sentiment signal regularises the stance head and supplies noisy but abundant sentiment gradients that drag the shared representations towards financially-meaningful directions. Once sentiment is close to ceiling, there is no room for the auxiliary task to hurt it.

b) Limitations.: (i) A 67%-class stance accuracy ceiling suggests sentence-level classification of FOMC prose is intrinsically limited–dovish/hawkish often resolves only at the document or paragraph level. (ii) We train 110M-parameter encoders; decoder-LLM approaches with longer context windows may improve stance. (iii) We did not perform data augmentation (paraphrasing, back-translation) which is a natural next step given the small FOMC corpus.

c) Broader impact.: A financial stance/sentiment classifier should be treated as *decision support* for analysts, not as the input to an autonomous trading strategy; misclassified FOMC sentences can and do occur on material macroeconomic news.

XI. Conclusion

We built twelve systems for financial sentiment and stance classification. Sentiment is effectively solved at 97%+ by any domain-pretrained or carefully fine-tuned transformer, while stance tops out at 0.6384 macro-F1 with our Multi-task FinBERT–tied by a BERT-base model with LLRD, gradual unfreezing and label smoothing. Multi-task learning gives the largest relative lift on the harder task, and domain pre-training remains most valuable in low-data regimes.

References

[1] T. Loughran and B. McDonald, "When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks," *The Journal of Finance*, vol. 66, no. 1, pp. 35–65, 2011.

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[3] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," *arXiv:1908.10063*, 2019.

[4] Y. Yang, M. C. S. Uy, and A. Huang, "FinBERT: A pretrained language model for financial communications," *arXiv:2006.08097*, 2020.

[5] A. Shah, S. Paturi, and S. Chava, "Trillion dollar words: A new financial dataset, task & market analysis," in *Proc. ACL*, 2023.

[6] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, and P. Takala, "Good debt or bad debt: Detecting semantic orientations in economic texts," *J. Assoc. Inf. Sci. Technol.*, vol. 65, no. 4, pp. 782–796, 2014.

[7] X. Liu, P. He, W. Chen, and J. Gao, "Multi-task deep neural networks for natural language understanding," in *Proc. ACL*, 2019.